Shubhadeep Mandal

81602 63797 shubhadeepmandal11@gmail.com shubhadeep-mandal Seronic2001

Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Projects

Negotiate, Then Solve: Multi-Agent Workflow Automation | *Python, Gemini, RAG, CP-SAT, FastAPI*

- Built a multi-agent pipeline (parsing, policy and negotiation agents) that turns free-text requests from email, chat and a web portal into typed constraints, resolving conflicts via CP-SAT and LLM negotiation through typed tool calls.
- Built a hybrid BM25 + dense-embedding RAG over OCR-ingested PDFs, scans and web pages, raising policy Recall@5 to **93.4%** (vs 87.7% BM25), with every denial required to cite a retrieved rule.
- Designed guardrails (authority checks, cited rules, human approval) so that **no attack succeeded** across 50 prompt-injection, spoofing and data-leak tests, even with a fully compromised parser.
- Shipped a FastAPI + React app backed by **215 pytest tests**; pooled Gemini keys with failover, rate limiting and a response cache, and LoRA fine-tuned Qwen3.5-4B on 2.7K distilled examples as a local fallback.

Retrieve Ten, Keep Three: Re-ranking RAG Evaluation | *Python, FAISS, Cross-Encoders, BM25, FastAPI*

- Built a two-stage RAG over the GDPR (bi-encoder + FAISS retrieves 10, cross-encoder keeps 3), raising Hit@3 from 0.46 to **0.85** on hand-labelled stress questions for **83 ms** per query (under 1% of end-to-end latency).
- Fused BM25 and dense retrieval with RRF, lifting Hit@10 from 0.77 to **0.93** on 100 generated questions for 1.3 ms; structure-aware chunking reached Hit@3 **0.79** on a quarter less LLM context.
- Deployed a FastAPI + React app on Railway with ONNX Runtime encoders to upload PDFs and compare dense, BM25 and fused ranks per chunk.

IndicLLM: Transformer Pretraining & Reasoning SFT | *PyTorch, SentencePiece, ONNX Runtime Web*

- Built Hindi and Assamese LMs end to end: curated 1.25B tokens (20%+ self-crawled and OCR'd), trained native 16K BPE tokenizers, and pretrained four 25M-param Transformers from PyTorch primitives on 500M tokens each.
- Fine-tuned on solver-generated reasoning, reaching **0.96** test accuracy and **0.85** on 10-hop chains (3× the training depth) vs a 0.48 baseline, with a leakage check ruling out answer lookup.
- Exported all 10 models to ONNX with a KV cache and a JavaScript tokenizer so they run fully in the browser, deployed as a live demo on Railway via Docker.

Peer-to-Peer Distributed File Sharing System | *C++, POSIX Sockets, Multithreading, OpenSSL, Linux*

- Built a BitTorrent-style P2P system in C++; self-scheduling workers cut 1GB single-peer downloads from **193s to under 13s**.
- Designed a pluggable piece-selection interface (rarest-first, sequential, random) chosen at runtime, with out-of-order parallel chunk writes to exact file offsets.
- Architected Primary-Backup tracker failover with race-free election and state sync; **46 unit tests** pass under ASan/UBSan.

Technical Skills

Languages: Python, C/C++, TypeScript, JavaScript, SQL, Bash.

Agentic AI & LLMs: Multi-Agent Systems, Tool Calling, RAG, Re-ranking, Guardrails, LoRA, Gemini API.

Machine Learning: PyTorch, Transformers, Hugging Face, SentencePiece, ONNX, FAISS, Scikit-learn, NumPy.

Software Engineering: Data Structures & Algorithms, OOP, FastAPI, React, pytest, Git, Docker, Linux, CI/CD.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

Interests

- Web Novel and Manga Enthusiast.
- Chess and Badminton.